Stochastic Processes: Generalization Theory

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Let us recall the definitions of some key concepts and clarify the notation. We denote $(\Omega, \mathcal{A}, \mathbb{P})$ as a probability space. A *random* or *stochastic process* is a collection of random variables $\{X_t : t \in T\}$ indexed by some set T. In most cases, the index set T is some subset of the real line, such as the natural numbers or an interval, giving the set T the interpretation of time. However, T can take more general forms, such as a metric space. In this manuscript, we will consider both cases. When T is an ordered set, we say that the random process $\{X_t\}_{t\in T}$ is adapted to a filtration $\{\mathcal{F}_t\}_{t\in T}$ when that the random variable X_t is measurable with respect to $\mathcal{F}_t$ where the filtration is formed by a monotone chain of sub-$\sigma-$algebras, i.e., $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{A}$ when $s \leq t$. The *natural* filtration is defined as the $\sigma-$algebra generated the previous random variables, i.e., $\mathcal{F}_t = \sigma(\{X_s\}_{s\leq t})$.

We will study the family of random variables whose tail behavior and concentration properties resemble those of a Gaussian distribution, i.e., the *sub-gaussian random variables.* The moment-generating function of a centered normal distribution $X \sim \mathcal{N}(0, \sigma^2)$ is

$$M_X(\lambda) = \mathbb{E}[e^{\lambda X}] = e^{\sigma^2\lambda^2/2}.$$

Subgaussian random variables are the class of random variables whose moment-generating function is bounded by that of a normal distribution. This leads to the following definition.

1 Subgaussian Random Variables

Definition 2 (Subgaussian random variable)**.** Let X be a random variable, we define the *log-moment generating function* ψ of X as

$$\psi(\lambda) := \log \mathbb{E}[e^{\lambda(X-\mathbb{E}[X])}]$$

and we say that X random variable is σ^2-subgaussian if its log-moment generating function satisfies $\psi(\lambda) \leq \lambda^2\sigma^2/2$ for all $\lambda \in \mathbb{R}$, and the smallest σ^2 for which that holds is called the variance proxy of X.

The following result is a variant of the Markov's inequality written in terms of the log-moment generating function.

Lemma 3 (Chernoff bound)**.** Let X be a σ^2-subgaussian random variable. Then, the following inequality holds

$$\mathbb{P}[X - \mathbb{E}X \geq a] \leq \exp\left\{\frac{-a^2}{2\sigma^2}\right\}.$$

In particular, $\mathbb{P}[|X - \mathbb{E}X| \geq a] \leq 2\exp\{-a^2/2\sigma^2\}$.

Proof. We exponentiate inside the probability before appliying Markov's inequality. For any $\lambda \geq 0$, we have

$$\mathbb{P}[X - \mathbb{E}X \geq a] = \mathbb{P}[e^{\lambda(X-\mathbb{E}X)} \geq e^{\lambda a}] \leq e^{-\lambda a}\mathbb{E}[e^{\lambda(X-\mathbb{E}X)}] = e^{\psi(\lambda)-\lambda a}.$$

Since this holds for every $\lambda \in \mathbb{R}$, we can choose λ to obtain the best bound. The optimal choice is $\lambda = \frac{a}{\sigma^2}$. Substituting this into the inequality and using the bound $\psi(\lambda) \leq \lambda^2\sigma^2/2$, we obtain

$$\mathbb{P}[X - \mathbb{E}X \geq a] \leq \exp\left\{\left(\frac{a}{\sigma^2}\right)^2\sigma^2/2 - \left(\frac{a}{\sigma^2}\right)a\right\} = \exp\left\{\frac{-a^2}{2\sigma^2}\right\}.$$

□

Lemma 4 (Maximal inequality)**.** Let $\{X_k\}_{1\leq k\leq n}$ be a collection of σ^2-subgaussian random variables with the same variance, satisfying $\mathbb{E}[X_k] = 0$ for all $k = 1, \ldots, n$. Then

$$\mathbb{E}\left[\sup_{1\leq k\leq n} X_k\right] \leq \sqrt{2\sigma^2\log n}.$$

Proof. Since $-\log x$ is convex, by Jensen's inequality, we have for any $\lambda > 0$

$$\begin{aligned}
\mathbb{E}\left[\sup_k X_k\right] &= \mathbb{E}\left[\frac{1}{\lambda}\log(e^{\lambda \sup_k X_k})\right] \leq \frac{1}{\lambda}\log \mathbb{E}[e^{\lambda \sup_k X_k}] \\
&\leq \frac{1}{\lambda}\log \sum_{1\leq k\leq n} \mathbb{E}[e^{\lambda X_k}] \leq \frac{1}{\lambda}\log(ne^{\psi(\lambda)}) \\
&= \frac{\log n + \psi(\lambda)}{\lambda} \leq \frac{\log n + \lambda^2\sigma^2/2}{\lambda}.
\end{aligned}$$

Since this holds for every $\lambda > 0$, we can now optimize over λ on the right hand side. Differentiating and setting it equal to zero, we obtain the minimum at $\lambda = \frac{\sqrt{2\log n}}{\sigma}$. Substituting this value into the equation, we obtain the desired result

$$\mathbb{E}\left[\sup_{1\leq k\leq n} X_k\right] \leq \sqrt{2\sigma^2\log n}.$$

□

Lemma 5 (Hoeffding lemma)**.** Let X be a random variable such that $a \leq X \leq b$ a.s. for some $a, b \in \mathbb{R}$. Then,

$$\mathbb{E}[e^{\lambda(X-\mathbb{E}[X])}] \leq e^{\lambda^2(b-a)^2/8}.$$

In other words, X is a $(b-a)^2/4$-subgaussian random variable.

Proof. Let us define the mean-centered random variable $Y := X - \mathbb{E}[X]$, then $\tilde{a} \leq Y \leq \tilde{b}$ where $\tilde{a} := a - \mathbb{E}[X]$ and $\tilde{b} := b - \mathbb{E}[X]$. By definition, log-moment generating function of X is given by $\psi(\lambda) = \log \mathbb{E}[e^{\lambda Y}]$, along with its derivatives

$$\psi'(\lambda) = \frac{\mathbb{E}[Ye^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]}, \qquad \psi''(\lambda) = \frac{\mathbb{E}[Y^2 e^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]} - \left(\frac{\mathbb{E}[Ye^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]}\right)^2.$$

We can interpret $\psi''(\lambda)$ as the variance of the random variable Y under another probability measure. We define anew probability measure $\mathbb{Q}$ by setting $d\mathbb{Q} := \frac{e^{\lambda Y}}{\mathbb{E}[e^{\lambda Y}]} d\mathbb{P}$. The definition is well-posed because the Radon-Nikodym derivative $\frac{e^{\lambda Y}}{\mathbb{E}[e^{\lambda Y}]}$ is positive and integrates to 1. From this, we deduce that

$$\mathbb{E}_{\mathbb{Q}}[g(Y)] = \frac{\mathbb{E}[g(Y)e^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]}.$$

Therefore,

$$\mathrm{Var}_{\mathbb{Q}}(Y) = \mathbb{E}_{\mathbb{Q}}[X^2] - (\mathbb{E}_{\mathbb{Q}}[Y])^2 = \frac{\mathbb{E}[Y^2 e^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]} - \left(\frac{\mathbb{E}[Ye^{\lambda Y}]}{\mathbb{E}[e^{\lambda Y}]}\right)^2 = \psi''(\lambda).$$

We can bound the variance of Y as follows

$$\mathrm{Var}_{\mathbb{Q}}(Y) = \mathrm{Var}_{\mathbb{Q}}(Y - z) \leq \mathbb{E}_{\mathbb{Q}}[(Y-z)^2].$$

Since z can be any real number, we let $z = (\tilde{a} + \tilde{b})/2$. Given that $\tilde{a} \leq Y \leq \tilde{b}$ a.s., we obtain

$$\mathbb{E}_{\mathbb{Q}}[(Y-z)^2] = \frac{1}{4}\mathbb{E}_{\mathbb{Q}}[(2Y - \tilde{a} - \tilde{b})^2] \leq \frac{(\tilde{b} - \tilde{a})^2}{4} = \frac{(b-a)^2}{4}.$$

Next, since $\psi(\lambda) \leq \lambda^2\sigma^2/2$, we observe that $\psi'(0) = \psi(0) = 0$. Using the bound that $\psi''(\lambda) \leq \frac{(b-a)^2}{4}$ along with the Fundamental Theorem of Calculus, we obtain

$$\psi(\lambda) = \int_0^\lambda \int_0^\mu \psi''(\rho) d\rho d\mu \leq \frac{\lambda^2(b-a)^2}{8}.$$

Moreover, X is a σ^2-sbugaussian variable with $\sigma^2 = (b-a)^2/4$. □

Lemma 6 (Azuma)**.** Let $\{X_k\}_{1\leq k\leq n}$ be a stochastic process adapted to the natural filtration $\{\mathcal{F}_k\}_{1\leq k\leq n}$, i.e., $\mathcal{F}_k = \sigma(X_1, \ldots, X_k)$. Assume that the random variables satisfy:

$$\mathbb{E}[e^{\lambda X_k} | \mathcal{F}_{k-1}] \leq e^{\lambda^2\sigma_k^2/2} \quad a.s. \qquad\qquad k = 1, \ldots, n \tag{1}$$

where σ_k^2 is the variance of X_k. Then, $\mathbb{E}[X_k] = 0$ and the sum $\sum_{k=1}^n X_k$ is a σ^2-subgaussian with variance proxy $\sigma^2 := \sum_{k=1}^n \sigma_k^2$.

Proof. The tower property of expectation assures that $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[Y|\mathcal{F}_{k-1}]]$. For any $1 \le k \le n$ the random variables $X_1, \ldots, X_{k-1}$ are $\mathcal{F}_{k-1}-$measurables. Then, it follows

$$\begin{aligned}\mathbb{E}[e^{\lambda \sum_{i=1}^{k} X_i}] = \mathbb{E}[\mathbb{E}[e^{\lambda \sum_{i=1}^{k-1} X_i} e^{\lambda X_k}|\mathcal{F}_{k-1}]] &= \mathbb{E}[e^{\lambda \sum_{i=1}^{k-1} X_i}\mathbb{E}[e^{\lambda X_k}|\mathcal{F}_{k-1}]] \\ &\overset{\text{eq. (1)}}{\le} e^{\lambda^2\sigma_k^2/2}\mathbb{E}[e^{\lambda \sum_{i=1}^{k-1} X_i}].\end{aligned}$$

By induction, it yields $\mathbb{E}[e^{\lambda \sum_{i=1}^{n} X_i}] \le e^{\lambda^2 \sum_{i=1}^{n} \sigma_i^2/2} = e^{\lambda^2\sigma^2/2}$. Therefore, assuming that $\mathbb{E}[X_k] = 0$, it follows

$$\psi(\lambda) = \log \mathbb{E}[e^{\lambda \sum_{i=1}^{n} X_i}] \le \lambda^2\sigma^2/2.$$

Let us prove that $\mathbb{E}[X_k|\mathcal{F}_{k-1}] = 0$ (which, by the tower property, implies that $\mathbb{E}[X_k] = 0$). We observe that equality holds in (1) when $\lambda = 0$. Differentiating both sides with respect to λ and evaluating at $\lambda = 0$ must yield the same result. This leads to the condition:

$$\frac{d}{d\lambda}\Big|_{\lambda=0} \mathbb{E}\left[e^{\lambda X_k}|\mathcal{F}_{k-1}\right] \overset{DCT}{=} \mathbb{E}\left[\frac{d}{d\lambda}\Big|_{\lambda=0} e^{\lambda X_k}|\mathcal{F}_{k-1}\right] = \frac{d}{d\lambda}\Big|_{\lambda=0} e^{\lambda^2\sigma_k^2/2} \Rightarrow \mathbb{E}[X_k|\mathcal{F}_{k-1}] = 0.$$

□

Corollary 8 (Azuma-Hoeffding inequality)**.** Let $\{\mathcal{F}_k\}_{1\le k\le n}$ be a filtration, and let X_k, A_k, B_k be stochastic processes satisfying that A_k, B_k are $\mathcal{F}_{k-1}-$measurable and $A_k \le X_k \le B_k$ a.s. for $k = 1, \ldots, n$. Then, $\sum_{k=1}^{n} X_k$ is σ^2-subgaussian with $\sigma^2 = \frac{1}{4}\sum_{k=1}^{n} \|B_k - A_k\|_\infty^2$. In particular, for every $t \ge 0$ the tail bound

$$\mathbb{P}\left[\sum_{k=1}^{n} X_k \ge t\right] \le \exp\left(-\frac{2t^2}{\sum_{k=1}^{n} \|B_k - A_k\|_\infty^2}\right).$$

Proof. Applying Hoeffding's Lemma 5 to X_k conditionally on $\mathcal{F}_{k-1}$ yields

$$\mathbb{E}[e^{\lambda X_k}|\mathcal{F}_{k-1}] \le e^{\lambda^2(B_k-A_k)^2/8}.$$

In other words, X_k is σ_k^2-subgaussian with $\sigma_k^2 = \frac{1}{4}(B_k - A_k)^2$. This follows because A_k and B_k are $\mathcal{F}_{k-1}$ measurable, allowing us to treat them as constants. On the other hand, applying Azuma's Lemma 6 to X_k implies that $\mathbb{E}[X_k] = 0$ and that $\sum_{k=1}^{n} X_k$ is σ^2-subgaussian with $\sigma^2 = \sum_{k=1}^{n} \sigma_k^2 = \sum_{k=1}^{n} \frac{1}{4}(B_k - A_k)^2$.

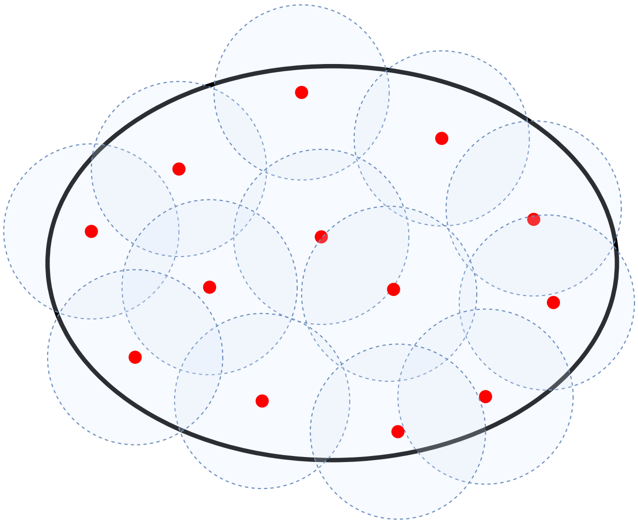
To establish the second part, note that $\mathbb{E}X_k = 0$, therefore, invoking Chernoff Bound Lemma 3 leads to the desired tail bound. □

2 Covering and Packing number

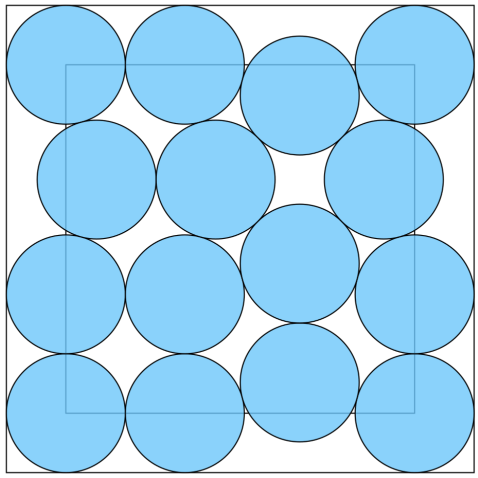
We will define the covering and packing number of a metric space (E, d). Intuitively, the *covering number* is the minimum number of balls required to completely cover the space, while the *packing number* is the maximum number of disjoint balls that can be placed within the space without overlapping (see Figure 1).

Definition 10 ($\epsilon-$net and covering number)**.** A set $N \subseteq E$ is called a $\epsilon-$*net* for (E, d) if for every $x \in E$, there exists $\pi(x) \in N$ such that $d(x, \pi(x)) \le \epsilon$. The smallest cardinality of an $\epsilon-$net for (E, d) is the called the *covering number*

$$N(E, d, \epsilon) := \inf\{|N| : N \text{ is an } \epsilon\text{−net for } (E, d)\}.$$



(a) The ellipse represents the T and the dots in red are the the elements in N, i.e., the $\epsilon-$net. The circles around the dots are the balls of ϵ radius that should cover the metric space.



(b) The optimal packing problem in a square using the blue balls.

Figure 1: Packing vs Covering number

Definition 12 ($\epsilon-$packing and packing number)**.** A set $N \subseteq E$ is called an $\epsilon-$*packing* of (E, d) if $d(x, x') > \epsilon$ for every $x, x' \in N$, $x \neq x'$. The largest cardinality of an $\epsilon-$packing of (E, d) is called the *packing number*

$$D(E, d, \epsilon) := \sup\{|N| : N \text{ is an } \epsilon-\text{packing of } (E, d).\}.$$

In the literature, the binary logarithm of the covering number of a set E is commonly referred to as the $\boldsymbol{\epsilon}$**−entropy** of E, denoted by

$$H_\epsilon(E) := \log_2 N(E, \|\cdot\|, \epsilon).$$

Similary, the binary logarithm of the packing number is typically called the $\boldsymbol{\epsilon}$**−capacity** of E, defined as

$$C_\epsilon(E) := \log_2 D(E, \|\cdot\|, \epsilon).$$

The logarithm is usually taken with base 2 because it used in the context of Computer Science to measure the transmission properties of certain channel.

The following lemma illustrates the relationship between the covering number and the packing number.

Lemma (Duality between covering and packing number)**.** For every $\epsilon > 0$

$$D(E, d, 2\epsilon) \leq N(E, d, \epsilon) \leq D(E, d, \epsilon).$$

Proof. (1) Let D be a $2\epsilon-$packing and let N be an $\epsilon-$net. For every $x \in D$, choose $\pi(x) \in N$ such that $d(x, \pi(x)) \leq \epsilon$. Then, for every $x' \in D$ such that $x \neq x'$, we have

$$2\epsilon < d(x, x') \leq d(x, \pi(x)) + d(\pi(x), \pi(x')) + d(\pi(x'), x') \leq 2\epsilon + d(\pi(x), \pi(x')),$$

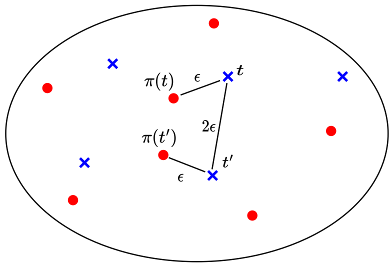


Figure 2: distance between x and x'.

which implies that $\pi(x) \neq \pi(x')$ (see Figure 2). Thus, the function $\pi : D \to N$ is injective, and thus, $|D| \leq |N|$. In other words, $D(E, d, 2\epsilon) \leq N(E, d, \epsilon)$.

(2) Let D be a *maximal* $\epsilon-$packing with $|D| = D(E, d, \epsilon)$. We claim that D is necessarily an $\epsilon-$net. Indeed, suppose for contradiction that there exists a point $x \in E$ such that $d(x, x') > \epsilon$ for every $x' \in D$. This would imply that $D \cup \{x\}$ is a *larger* $\epsilon-$packing, contradicting the maximality of D. Therefore, every point in E must be within a distance at most of ϵ from some point in D, confirming that D is an $\epsilon-$net. □

We are now ready to establish an upper bound on the covering number of the Euclidean ball B_2^n with respect to the Euclidean distance. The proof of this fundamental result employs a clever technique known as a *volume argument*.

Lemma. Let B_2^n be the $n-$dimensional euclidean ball centered at zero with radius 1, i.e., $B_2^n := \{x \in \mathbb{R}^n : \|x\|_2 < 1\}$. Then, we have that $N(B_2^n, \|\cdot\|, \epsilon) = 1$ for $\epsilon \geq 1$ and

$$\left(\frac{1}{\epsilon}\right)^n \leq N(B_2^n, \|\cdot\|, \epsilon) \leq \left(\frac{3}{\epsilon}\right)^n \qquad \text{for } 0 < \epsilon < 1.$$

Proof. Case $\epsilon \geq 1$: It is clear that for $\epsilon \geq 1$, $N(B_2^n, \|\cdot\|, \epsilon) = 1$ since $N := \{0\}$ is a $\epsilon-$net, i.e., $\|x - 0\| < 1 \leq \epsilon$ for every $x \in B_2^n$.

Case $0 < \epsilon < 1$: Let us begin with the upper bound. Let D be a $2\epsilon-$packing of B_2^n. Since $d(x, x') > 2\epsilon$ for all $x \neq x'$ in D, the balls $\{B(x, \epsilon) : x \in D\}$ must be disjoint. On the other hand, every ball $B(x, \epsilon)$ for $x \in B_2^n$ must be cointained in a larger ball centered in zero with radius $1 + \epsilon$, i.e., $B(x, \epsilon) \subseteq B(0, 1 + \epsilon)$. Therefore, the sum of volumes of the balls satisfy

$$\sum_{x \in D} \lambda(B(x, \epsilon)) = \lambda\left(\bigcup_{x \in D} B(x, \epsilon)\right) \leq \lambda(B(0, 1 + \epsilon))$$

where λ denotes the Lebesgue measure on $\mathbb{R}^n$. Using that the Lebesgue measure is homogeneous and that $\lambda(B(0, \alpha)) = \lambda(\alpha B(0, 1)) = \alpha^n \lambda(B(0, 1))$, then

$$\epsilon^n |D| B(0, 1) = |D| B(0, \epsilon) = \sum_{x \in D} \lambda(B(x, \epsilon)) \leq \lambda(B(0, 1 + \epsilon)) = (1 + \epsilon)^n B(0, 1).$$

Therefore, dividing both sides by $B(0, 1)$, we obtain that

$$|D| \leq \left(\frac{1 + \epsilon}{\epsilon}\right)^n.$$

We have established that for every $2\epsilon-$packing D, we obtain an upper bound for $D(E, d, 2\epsilon)$. This leads to the following chain of inequalities

$$N(B_2^n, \|\cdot\|, 2\epsilon) \overset{\text{Lemma 2}}{\leq} D(B_2^n, \|\cdot\|, 2\epsilon) \leq \left(1 + \frac{1}{\epsilon}\right)^n \leq \left(\frac{3}{2\epsilon}\right)^n \qquad \text{for } 2\epsilon < 1.$$

Relabeling 2ϵ as ϵ completes the proof.

We proceed similarly to obtain the lower bound. Let N be an $\epsilon-$net for B_2^n. Then,

$$\lambda(B_2^n) \leq \lambda\left(\bigcup_{x\in N} B(x,\epsilon)\right) \leq \sum_{x\in N} \lambda(B(x,\epsilon)) = |N|\epsilon^n \lambda(B_2^n).$$

Hence, dividing by $\lambda(B_2^n)$, we obtain

$$|N| \geq \frac{\lambda(B_2^n)}{\lambda(B(0,\epsilon))} = \left(\frac{1}{\epsilon}\right)^n.$$

This inequality holds for every $\epsilon-$net N, so we conclude that $N(B_2^n, \|\cdot\|, \epsilon) \geq (1/\epsilon)^n$. □

The following Lemma finds a bound for the growing rate of the $\epsilon-$entropy of the Sobolev space H^{m+1}. The proof can be found in [Nickl and Pötscher, 2007, Corollary 4].

Lemma (ϵ-Entropy of $H^{m+1}(\Omega, \mathbb{R}^{d_2})$)**.** Let $\Omega \subseteq \mathbb{R}^{d_2}$ be a Lipschitz domain. For $m \geq 1$, one has

$$\log N(B_{H^{m+1}(\Omega)}(1), \|\cdot\|_{H^{m+1}(\Omega)}, \epsilon) = \mathcal{O}_{\epsilon\to 0}(\epsilon^{-d_1/(m+1)}).$$

3 Subgaussian Process

Definition 14 (Subgaussian process)**.** A random process $\{X_t\}_{t\in T}$ on a metric space (T, d) is called *subgaussian* if $\mathbb{E}[X_t] = 0$ for all $t \in T$ and

$$\mathbb{E}[e^{\lambda(X_t - X_s)}] \leq e^{\lambda^2 d(t,s)^2/2} \qquad \text{for all } t, s \in T, \lambda \geq 0. \tag{2}$$

Notice that for every $s, t \in T$, the random variable $X_t - X_s$ is a $d(t,s)^2$-subgaussian random variable.

Definition (Separable process)**.** A random process $\{X_t\}_{t\in T}$ is called separable if there exists a countable set $T_0 \subseteq T$ such that

$$X_t \in \lim_{\substack{s\to t\\ s\in T_0}} X_s \qquad \text{for all } t \in T \quad a.s.$$

Here, $\lim_{\substack{s\to t\\ s\in T_0}} X_s \equiv \{X_t : \exists \{s_n\}_{n\in\mathbb{N}} \text{ such that } X_{s_n} \xrightarrow{n\to\infty} X_t \text{ a.s.}\}$

Remark. The assumption of separability is almost always satisfied. For example, assuming that $t \to X_t$ is continuous and T is a separable metric space (as is the case in this manuscript), then we can take T_0 to be any countable dense subset of T, allowing us to verify the separability assumption.

For the next theorem, we must prove that $\sup_{t\in T} X_t$ is measurable. In fact, if T is uncountable, the supremum is not necessarily measurable. However, under the assumption of separability, we have $\sup_{t\in T} X_t = \sup_{t\in T_0} X_t$. Since the supremum of a measurable function is always measurable, it follows that $\sup_{t\in T} X_t$ must be measurable.

Theorem 15 (Dudley)**.** Let $\{X_t\}_{t\in T}$ be a separable subgaussian process in the metric space (T, d). Then,

$$\mathbb{E}\left[\sup_{t\in T} X_t\right] \leq 6 \sum_{k\in\mathbb{Z}} 2^{-k} \sqrt{\log N(T, d, 2^{-k})}.$$

Proof. We begin by proving the result for the case where $|T| < \infty$. Let $k_0 \in \mathbb{Z}$ be the largest integer such that $2^{-k_0} \geq \text{diam}(T)$. It is clear that for every $t_0 \in T$, the set $N_0 := \{t_0\}$ forms a $2^{-k_0}-$net and $\pi_0(t) \equiv t_0$.

For $k > k_0$, let N_k be a $2^{-k}-$net such that $|N_k| = N(T, d, 2^{-k})$. We denote $\pi_k(t)$ as the element in N_k that satisfies $d(t, \pi_k(t)) \leq 2^{-k}$. Using a chaining argument up to the scale 2^{-n}, we proceed as follows

$$\begin{aligned}\mathbb{E}\left[\sup_{t\in T} X_t\right] &= \mathbb{E}\left[\sup_{t\in T}\left\{X_{\pi_0(t)} + \left(\sum_{k=k_0+1}^{n} X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\right) + X_t - X_{\pi_n}(t)\right\}\right] \\ &\leq \mathbb{E}[X_{t_0}] + \sum_{k=k_0+1}^{n} \mathbb{E}\left[\sup_{t\in T}\{X_{\pi_k(t)} - X_{\pi_{k-1}(t)}\}\right] + \mathbb{E}\left[\sup_{t\in T}\{X_t - X_{\pi_n(t)}\}\right].\end{aligned}$$

By definition of subgaussian process, $\mathbb{E}[X_{t_0}] = 0$. Since $|T| < \infty$, we can choose n sufficiently large so that $N_n = T$, and hence $\pi_n(t) = t$, meaning that the third term vanishes. Next, we bound the second term. By definition, $X_{\pi_k(t)} - X_{\pi_{k-1}}(t)$ is a $d(\pi_k(t), \pi_{k-1}(t))$-subgaussian random variable. We can readily estimate the variance,

$$d(\pi_k(t), \pi_{k-1}(t)) \leq d(\pi_k(t), t) + d(t, \pi_{k-1}(t)) \leq 2^{-k} + 2^{-(k-1)} = 3 \times 2^{-k}.$$

Moreover, we can control the number of terms in the sum, note that $\{X_{\pi_k(t)} - X_{\pi_{k-1}(t)} : t \in T\}$ contains at most $|N_k||N_{k-1}|$ which is bounded by $|N_k|^2$ terms. Applying Maximal Inequality Lemma 4 to these terms, we obtain

$$\begin{aligned}\mathbb{E}\left[\sup_{t\in T} X_t\right] &\leq \sum_{k=k_0+1}^{n} \sqrt{2d(\pi_k(t), \pi_{k-1}(t))^2 \log|N_k|^2} \leq 6 \sum_{k=k_0+1}^{n} 2^{-k}\sqrt{\log|N_k|} \\ &\leq 6 \sum_{k=k_0+1}^{n} \sqrt{\log N(T, d, 2^{-k})}.\end{aligned}$$

To prove the result when $|T|$ is infinite, we can use separability to approximate the general case by the finite case. Indeed, by separability, there is a countable set $T \subseteq T'$ such that $\sup_{t\in T} X_t = \sup_{t\in T'} X_t$ a.s. Let us denote T'_k the first k elements of T'. Then, by monote convergence of expectations

$$\mathbb{E}\left[\sup_{t\in T} X_t\right] = \mathbb{E}\left[\sup_{t\in T'} X_t\right] = \mathbb{E}\left[\sup_{k\geq 1}\sup_{t\in T'_k} X_t\right] = \sup_{k\geq 1}\mathbb{E}\left[\sup_{t\in T'_k} X_t\right].$$

And, applying the chaining inequality to each finite maximum and using that $N(T_k, d, \epsilon) \leq N(T, d, \epsilon)$ yields the same result. □

Corollary (Entropy integral)**.** Let $\{X_t\}_{t\in T}$ be a separable subgaussian process on the metric space (T, d). Then,

$$\mathbb{E}\left[\sup_{t\in T} X_t\right] \leq 12 \int_0^\infty \sqrt{\log N(T, d, \epsilon)} d\epsilon.$$

Proof. Since $N(T, d, \cdot)$ is decreasing, we obtain the following chains of inequalities

$$\begin{aligned}
\sum_{k\in\mathbb{Z}} 2^{-k}\sqrt{\log N(T,d,2^{-k})} &= 2\sum_{k\in\mathbb{Z}}\int_{2^{-k-1}}^{2^{-k}} \sqrt{\log N(T,d,2^{-k})}d\epsilon \\
&\leq 2\sum_{k\in\mathbb{Z}}\int_{2^{-k-1}}^{2^{-k}} \sqrt{\log N(T,d,\epsilon)}d\epsilon \\
&= 2\int_0^{\infty} \sqrt{\log N(T,d,\epsilon)}d\epsilon.
\end{aligned}$$

□

Definition 17 (Discrete derivative)**.** Let $f \in C(\mathbb{R}^n, \mathbb{R})$. We define the *discrete derivative* of f with respect to variable x_k at the point $x \in \mathbb{R}$ as follows:

$$\mathfrak{D}_k f(x) := \sup_z f(x_1, \ldots, x_{k-1}, z, x_{k+1}, \ldots, x_n) - \inf_z f(x_1, \ldots, x_{k-1}, z, x_{k+1}, \ldots, x_n).$$

Theorem 18 (McDiarmid)**.** Let $X_1, \ldots, X_n$ be independent random variables, $f \in C(\mathbb{R}^n, \mathbb{R})$. Then, $f(X_1, \ldots, X_n)$ is σ^2-subgaussian with $\sigma^2 = \frac{1}{4}\sum_{k=1}^n \|\mathfrak{D}_k f\|_\infty^2$ where $\mathfrak{D}_k f$ is the discrete derivative (see Definition 17).

Proof. Let us define the following random processes for $k = 1, \ldots, n$

$$Y_k := \mathbb{E}[f(X_1, \ldots, X_n)|\mathcal{F}_k] - \mathbb{E}[f(X_1, \ldots, X_n)|\mathcal{F}_{k-1}].$$

where $\mathcal{F}_k$ is the natural filtration. We observe that it satisfies $\mathbb{E}[Y_k|\mathcal{F}_{k-1}] = 0$ due to the tower property. Moreover, we see that

$$\sum_{k=1}^n Y_k = f(X_1, \ldots, X_n) - \mathbb{E}f(X_1, \ldots, X_n). \tag{3}$$

On the other hand, let us denote $g_k(z) := f(X_1, \ldots, X_{k-1}, z, X_{k+1}, \ldots, X_n)$. Since X_k is indpendent from $X_1, \ldots, X_{k-1}, X_{k+1}, \ldots, X_n$, it follows that

$$\mathbb{E}[\inf_z g_k(z)|\mathcal{F}_{k-1}] = \mathbb{E}[\inf_z g_k(z)|\mathcal{F}_k] \leq \mathbb{E}[g_k(X_k)|\mathcal{F}_k] \leq \mathbb{E}[\sup_z g_k(z)|\mathcal{F}_k] = \mathbb{E}[\sup_z g_k(z)|\mathcal{F}_{k-1}].$$

Hence, $A_k \leq Y_k \leq B_k$ a.s. where

$$\begin{aligned}
A_k &:= \mathbb{E}[\inf_z g_k(z) - g_k(X_k)|\mathcal{F}_{k-1}], \\
B_k &:= \mathbb{E}[\sup_z g_k(z) - g_k(X_k)|\mathcal{F}_{k-1}].
\end{aligned}$$

We notice that A_k, B_k are $\mathcal{F}_{k-1}-$measurables. Then, the Azuma-Hoeffding inequality (Corollary 8) assures us that $\sum_{k=1}^n Y_k$ is σ^2-subgaussian with $\sigma^2 = \frac{1}{4}\sum_{k=1}^n \|B_k - A_k\|_\infty^2$. Moreover, by the above identity (3), $f(X_1, \ldots, X_n)$ is also a σ^2-subgaussian. We note that

$$|B_k - A_k| = |\mathbb{E}[\mathfrak{D}_k f(X)|X_1, \ldots, X_{k-1}]| \leq \|\mathfrak{D}_k f\|_\infty.$$

Therefore, $f(X_1, \ldots, X_n)$ is σ^2-subgaussian with $\sigma^2 = \frac{1}{4}\sum_{k=1}^n \|D_k f\|_\infty^2$.

□

References

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